

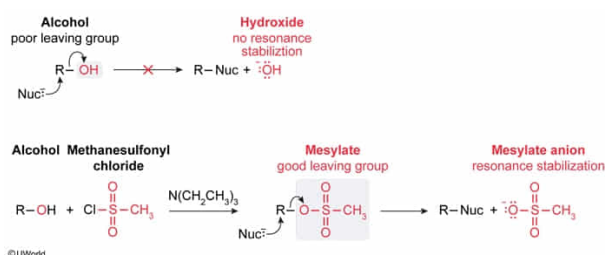
If an alcohol were to undergo a substitution reaction, which of the following modifications would improve the alcohol's leaving group ability?

- ☐ A. Protection of the alcohol [6%]
- ☐ B. Deprotonation of the alcohol [13%]
- ✓ ☒ C. Conversion of the alcohol to a mesylate [67%]
- ☐ D. Oxidation of the alcohol [11%]

Correct

68% answered correctly

### Explanation:



Substitution reactions require strong **nucleophiles and electrophiles** and good **leaving groups**. When a nucleophile attacks an electrophile, a bond is formed between the nucleophile and electrophile whereas the bond between the electrophile and the leaving group is broken. A leaving group containing an **electron withdrawing group** can enhance the partial positive charge of the electrophile. Leaving groups are weak bases and are able to accept electrons; they often can be stabilized by the delocalization of electrons through **resonance** and an **inductive effect**.

**Alcohols** are weak electrophiles and **poor leaving groups**; therefore, nucleophiles are unlikely to displace the hydroxyl group ( $-\text{OH}$ ). The alcohol can be converted into a **mesylate** ( $-\text{SO}_3\text{CH}_3$ ) to **improve its leaving group ability**. Mesylates are **produced by** the reaction of an alcohol with **methanesulfonyl chloride and a base**, such as triethylamine, and are good leaving groups because the **mesylate anion** can stabilize the negative charge through resonance. Similar to mesylates, **tosylates** ( $-\text{SO}_3\text{C}_6\text{H}_4\text{CH}_3$ ) can also be used to make an alcohol a better leaving group.

**(Choice A)** Protecting the alcohol does not make it a better leaving group but rather decreases its reactivity.

**(Choice B)** Deprotonation of the alcohol creates a negative charge on the alcohol oxygen atom, making it a nucleophile rather than a leaving group. Alternatively, protonation of the alcohol would create a strong electrophile and good **leaving group in the form of water**.

(Choice D) Oxidation of the alcohol forms a double bond to the oxygen atom. It makes the carbon bonded to the alcohol oxygen a strong electrophile that is susceptible to nucleophilic attack, but the alcohol leaving group ability is decreased as there are now two bonds holding it in place.

**Educational objective:**

Alcohols are poor leaving groups and weak electrophiles. However, an alcohol's leaving group ability can be improved through the formation of a mesylate ( $-\text{SO}_3\text{CH}_3$ ) or a tosylate ( $-\text{SO}_3\text{C}_6\text{H}_4\text{CH}_3$ ). Mesylates and tosylates can be produced by the reaction of an alcohol with a base and methanesulfonyl chloride or *p*-toluenesulfonyl chloride, respectively. The mesylate and tosylate anions can stabilize their negative charges by the delocalization of electrons through resonance.

**Time spent:**0 sec

**Subject:**

Organic Chemistry

**QID:**401497

**System:**

Functional Groups and Their Reactions